

SIGCHI Lifetime Practice Award Talk: Usability Barriers and How to Overcome Them

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ABSTRACT

How long is a lifetime of practice in human-computer interaction? In my case, it has been 42 years, almost all of that in IBM Research (with a couple of sabbatical years in the then new Watson Group). Having queued up for access to DEC and Prime machines during my PhD work in the mid 70s, I was positively thrilled when, upon arriving at the T.J. Watson Research Center in 1978, I found a dedicated 3277 “green screen” terminal on my desk attached to a 370 mainframe. Hot-rodded with an onboard character buffer and double speed cursor circuitry, I could fly through writing and coding, personally experiencing the productivity benefits of sub-second response time [1].

Beyond some great hardware (for the time), I also had extraordinary colleagues. I imprinted on the working style of John Gould and Stephen Boies as we built the second hardware iteration of what became the world’s first commercial voice mail system and its final manifestation as the messaging system for the 1984 Summer Olympics. In this talk I’ll share some of the still relevant behavioral principles of system design we learned in the course of that project [2].

Beyond this, I’ll review work in early (pre-Web!) Internet access for schools [3], home shopping [4], Web accessibility [5], and programmer productivity in peta-scale scientific computing [6]. Each project will highlight a barrier to building useful and usable systems, barriers that still persist. More importantly, I’ll tell you how we overcame those barriers, giving you something that will, perhaps, be useful in your own work.

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1 BIO

John T. Richards received his PhD in Cognitive Psychology from the University of Oregon in 1978. He joined the IBM Research Division in that same year as part of the team creating the world’s first voice mail system. Incorporating one of the earliest instances of a user interface management system, this work was recognized by the Human Factors Society Alexander C. Williams Jr. Award for “outstanding human factors contributions to the design of a major operational system”. Following this he helped form and subsequently led IBM’s User Interface Institute, a group noted for research in both usability engineering and object-oriented programming. Returning to the user-centered design and development of novel applications after several years, John and his team created: NetVista, a bespoke suite of fully autonomic servers and multi-platform clients enabling K-12 schools to easily and safely access the Internet; SmartPad, IBM’s first mobile e-business application pioneering the concept of retail shopping via a mobile device; and accessibilityWorks, a multi award winning browser extension allowing people with visual, motor, and cognitive disabilities to access the World Wide Web.

John is currently an IBM Distinguished Research Staff Member and has served for several years as an Honorary Professor in the School of Engineering at the University of Dundee, Scotland. Recent scientific contributions include the development of an empirical and behavioral-modeling approach to quantifying human productivity in parallel and scientific computing (funded by DARPA’s High Productivity Computing Systems initiative). Recent design and development contributions include the creation of Web experiences illustrating the use of Fairness and Explainability toolkits in furtherance of Trustworthy AI. John is an ACM Fellow, a Fellow of the British Computer Society, a recipient of ACM SIGPLAN’s Distinguished Service Award, a member of IBM’s Academy of Technology, and a recipient of multiple Outstanding Level awards from IBM.

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